ESP32 BLE Broadcast - Simple PRD

Objective

Create firmware for an ESP32 DevKit that starts BLE advertising automatically after boot.

BLE Device Name

The BLE advertised device name shall be generated dynamically in the format:

`FirmGen_XX`

XX shall be the last 2 hexadecimal digits of the ESP32 MAC address.

Example: If the MAC address is A4:CF:12:34:56:9B, the BLE name shall be FirmGen_9B.

Requirements

1. Initialize BLE after device boot.
2. Read the ESP32 MAC address at runtime.
3. Extract the last 2 hexadecimal digits of the MAC address.
4. Generate the BLE name as FirmGen_XX.
5. Start BLE advertising using the generated name.
6. Keep BLE advertising continuously.

UART Output

Use UART at 115200 baud for basic debug logs.

Print the detected MAC address.

Print the generated BLE device name.

Print a message when BLE advertising starts.

Acceptance Criteria

The ESP32 shall boot successfully and start BLE advertising.

The advertised name shall match FirmGen_ followed by the last 2 digits of the MAC address.

The device shall be visible from a BLE scanner or mobile BLE application.